

Junyi (Conny) Zhou

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EDUCATION

- Harvard University** Boston, MA
M.S. in Health Data Science August 2025 – March 2027
 - Program requirements complete December 2026 (commencement March 2027); available for full-time start January 2027
 - Coursework: Deep Learning, AI in Medicine, Healthcare Machine Learning, Data Science
- Emory University, College of Arts and Sciences** Atlanta, GA
B.S. in Applied Mathematics and Statistics; Minor in Computer Informatics; GPA: 3.925 August 2021 – May 2025
 - Coursework: Probability and Statistics, Machine Learning, NLP, Data Structures, Database Systems

TECHNICAL SKILLS

- Languages:** Python, C++, Java, SQL, R, JavaScript
- ML & Deep Learning:** PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, Hugging Face Transformers, scanpy, AnnData
- LLM & Applied AI:** RAG, agentic pipelines, prompt engineering, embeddings & vector search, OpenAI API, MongoDB Atlas Vector Search
- Backend & DevOps:** Flask, REST APIs, React, Docker, Auth0, Git, YAML policy compilers
- Data & Compute:** PostgreSQL, MySQL, MongoDB, Spark, Hadoop, SLURM/HPC, FASRC Cannon, job arrays
- Cloud:** AWS (SageMaker, Lambda, EC2, S3, Redshift, Elastic Beanstalk, CloudFront), Snowflake, BigQuery
- Certifications:** AWS Certified Cloud Practitioner, AWS Certified AI Practitioner, AWS Machine Learning Associate

RESEARCH EXPERIENCE

- speciesOT: Cross-Species Single-Cell Translation** [GitHub] Boston, MA
Graduate Researcher — Mooney Lab & Alvarez-Melis Lab, Wyss Institute February 2026 – Present
 - Designed a mouse-to-human single-cell transport that compares an **ICNN optimal-transport** map (IMPACT_CellOT) with an **scGen** latent mean-shift, both inside one shared autoencoder. The treatment-composition leg is specified and not built.
 - Built a **hub CLI** that catalogs trained runs, ranks them on decoded-space gap-closed rather than raw R^2 , and prints **SLURM** chains. The hub never submits jobs.
 - Engineered atlas-scale handling: backed AnnData prep so the documented **43 GB** Tabula files are not full-loaded on a login node, and baked gene-axis plus scGen-shift sidecars so new mouse prediction does not reopen the training atlas.
 - On a frozen paired ablation, scGen-split R^2 was **0.85–0.90** with the source species in training versus **0.65–0.67** on the held-out-species split, and gene-space ranking reversed with split (6,619 orthologs vs 1,000 HVGs).
- Structure-Verified RLVR for Label-Efficient Pathology Instance Segmentation** [GitHub] Boston, MA
Graduate Researcher — Harvard Health Data Science capstone (mentor Alexander Chowdhury) August 2026 – Present
 - Specified **S-Seg-RLVR**, a capstone that uses verifiable rewards (instance count, separation, topology) instead of only Dice/IOU, with a planned SAM/SAM2 policy and composite GRPO reward.
 - Stood up the public repo: proposal, a 30-paper literature library, a 16-week roadmap, and typed reward interfaces. Training and datasets are next. No segmentation metrics yet.
- AlphaFold Protein–Nucleic Acid Interaction Prediction** [GitHub] Atlanta, GA
Data Analyst — Bou-Nader Lab, Emory Dept. of Biochemistry February 2025 – August 2025
 - Built a pipeline that runs **AlphaFold-Multimer** and **AlphaFold3** on **MSA** inputs and ranks structures by **pLDDT** and **PAE**.
 - Scheduled GPU inference on shared **HPC** with **SLURM** and shipped a documented CLI so non-ML researchers could run and triage jobs.
- Computational Legislative Studies: UK & New Zealand Parliamentary Data** [GitHub] Atlanta, GA
Research Assistant — Emory University (Carrubba & Estrada) December 2023 – May 2024
 - Built scrapers and **ETL** over UK Parliament open data, normalizing 22,606 member-level division votes, 4,791 MPs, 4,108 bills, and 3,875 constituencies.
 - Diagnosed **Cloudflare** 403 challenges and restored collection with TLS-impersonating requests and **Selenium**, then handed the pipeline off for a UK-to-New-Zealand continuation.

ENGINEERING PROJECTS

- **Closed-Loop Cluster Autoresearch for scGen vs CellOT** [GitHub] Boston, MA
Independent Project — Harvard FASRC Cannon Cluster June 2026 – Present
 - Built a self-directed experiment loop (submit → watch → reflect → decide → synthesize) that ran **338** SLURM jobs (318 clean, 20 handled errors) over **79.3 hours**. The public ledger is a header only; those counts come from the 2026-07-29 Cannon working tree and the README.
 - Split a deterministic one-delta proposer from the cluster substrate and **checkpointed each agenda to disk**, so the cluster keeps the last plan if the planner is offline.
 - Routed tiny-MLP training to CPU partitions because GPU carries 200–500× fairshare cost, and made runs preemption-tolerant on `serial_requeue` via checkpoint/resume and a capped job array.
 - Implemented an optional **LLM director** with token and cost accounting and a spend ceiling that falls back to the rule reasoner. No usage ledger exists. Historical episodes used the rule reasoner.
- **mixhvg-py: Validated Python Port of an R HVG Ensemble** [GitHub] Boston, MA
Independent Project — Scientific software July 2026
 - Ported **mixhvg** to an **AnnData**-native API (cells × genes), combining methods by **maximum rank** rather than averaging.
 - Validated against R 4.3.3 / scran 1.30.0 / Seurat 5.3.0: Seurat vst and dispersion match at Spearman **1.0000**; the paper-recommended mixture agrees at Jaccard **0.980** (495 of 500 genes).
 - Found that a weighted-mean log-space correction overshot scran’s trend by about 3.4% and raised discarded genes from 0.499 to 0.719. A weighted median restored 0.503 vs R 0.499.
 - Reproduced two degenerate upstream methods and a **Seurat 5** silent-NULL vst bug in tests rather than silently “fixing” them. 68 pytest tests pass.
- **Job Search OS: Human-Supervised Browser Automation** [GitHub] Boston, MA
Independent Project July 2026 – Present
 - Built a job-search operating system that keeps **GitHub** as durable policy and evidence, a **Google Sheet** as Polar runtime state, and **Simplify** as the application ledger.
 - Compiled Polar hourly workflows from repo YAML so the browser agent reads one runtime file. Polar may Submit only after duplicate, identity, and writing gates. Cloud Submit at G2/G3 stays closed.
 - Encoded the policy in 16 knowledge YAML files and 10 Polar workflow specs, guarded by 20 unittest files (220 tests) covering runtime compile, queue writeback, and Computer Use action sheets.
- **Airway Management Simulation Chatbot** [Live Demo] Atlanta, GA
Full-Stack Engineer, Scrum Master — Emory Pediatric Hospital January 2024 – August 2025
 - Architected a clinical **agentic LLM** pipeline pairing **RAG** over **MongoDB Atlas Vector Search** with prompt and context-window management to generate multi-turn training scenarios from **OpenAI** models.
 - Built a HIPAA-conscious full-stack platform (**Python/Flask** REST API, React, MongoDB, **Auth0**) on **AWS** Elastic Beanstalk and CloudFront.
 - Led an Agile team of 6, running code reviews and folding clinician feedback into successive releases.